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ChooChooTracker

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ChooChooTracker is an eight-track music tracker designed for small screens and handheld consoles. Its workflow follows the LSDj family of trackers: a song contains chains, chains contain phrases, and phrases contain notes, instruments and FX. Each instrument chooses its own sound engine, so AY, Braids, Plaits and PCM samples can be freely mixed in one project.

This manual describes the current development build. Features marked as requiring hardware validation should be tested on the target ArkOS/PortMaster console before live use.

1. Installation and files

ArkOS / PortMaster

Install `choochootracker.zip` through PortMaster, or copy the extracted package to the console's `ports` directory. Start **ChooChooTracker** from the Ports menu.

Projects use the `.cct` extension. Instrument files use `.cni`. PCM instruments load uncompressed WAV files. Keep projects, instruments and samples in their configured folders; a PCM project stores the sample path, not a second copy of the audio file.

Windows

Run `choochootracker.exe` with `SDL2.dll` and `libwinpthread-1.dll` in the same directory.

2. Controls

ChooChooTracker uses eight logical controls.

Logical control	Handheld default	Windows default
Left, Right, Up, Down	D-pad	Arrow keys
Edit	A	X
Opt	B	Z
Play	Start	Space
Select	Select	Shift

Mappings can be changed in **Settings > Key mapping**.

Navigation

- Use the D-pad to move the cursor.
- Hold **Select** and press a direction to move between screens.
- Hold **Opt** and press a direction for screen-specific navigation.
- On a value, hold **Edit** and press Left/Right for fine changes or Up/Down for coarse changes.
- Tap **Edit** to enter a value or activate a command.
- Double-tap **Edit** to create a new chain, phrase or instrument where supported.
- **Opt + Edit** clears or cuts the current value.
- Holding a direction repeats after the delay configured in Settings.

Selection and clipboard

- **Select + Opt** enters selection mode and cycles through useful selection ranges.
- Move the D-pad to extend the range.
- **Opt** copies the selection.
- **Opt + Edit** cuts it.
- **Select + Edit** pastes it.
- **Edit + direction** edits the selected cells together.

Playback

- **Play** starts from the cursor; press it again to stop.
- **Select + Play** starts all tracks when used outside the Song screen.
- **Edit + Play** previews an instrument from the Instrument Pool.

3. Song structure and screen map

The main row is:

M S C P I T

- **M**: Mixer
- **S**: Song
- **C**: Chain
- **P**: Phrase
- **I**: Instrument
- **T**: Table

Secondary screens appear above or below their parent. Project and Settings surround Song. Reverb and Delay surround Mixer. Groove belongs to Phrase; Instrument Pool and Modulation belong to Instrument; Wavetable belongs to Table.

The hierarchy is:

Song track -> Chain -> Phrase row -> Note + Instrument + FX

There are always eight independent tracks. An AY instrument used on one track does not share a forced three-channel chip with other tracks.

4. A first pattern

1. Open the Song screen and create or enter a chain number on a track.
2. Move to Chain and enter a phrase number.
3. Move to Phrase and enter notes.
4. Enter an instrument number next to a note. Create that instrument in the Instrument screen or Instrument Pool.

5. Press Play.
6. Add FX in the phrase FX columns as needed.
7. Set track levels and sends in Mixer, then save the project from Project.

5. Mixer, Reverb and Delay

The Mixer is the leftmost main screen. Its CPU display reports audio callback load; it is useful for checking a project on the actual handheld.

Each track has:

Field	Meaning
LVL	Post-engine track level, 000–100
REV	Send to the shared Clouds reverb, 000–100
DLY	Send to the shared ping-pong delay, 000–100
MUTE	Silence this track
SOLO	Listen to this track alone

The mixer is track-based because it controls the eight playback lanes. If a track changes instruments, its level and sends remain attached to the track.

Clouds Reverb

Press **Select + Up** from Mixer to open Reverb; repeat it to return.

- **Return:** wet reverb level in the master mix.
- **Time:** decay/time control, 00–FF .
- **Damping:** high-frequency absorption, 00–FF .
- **Filter:** low-pass filter applied before the reverb.

This is the reverb algorithm from Mutable Instruments Clouds, adapted as a shared send effect. It is not the complete Clouds granular processor.

Ping-pong Delay

Press **Select + Down** from Mixer to open Delay; repeat it to return.

- **Return:** wet delay level.
- **Ticks:** delay time in tracker ticks.
- **Feedback:** cross-feedback amount, limited to 95%.
- **Filter:** low-pass filter in the delayed signal path.

The first repeat alternates according to the stereo input and subsequent feedback crosses between left and right.

6. Song, Chain and Phrase

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Song

The Song screen contains eight columns, one per track. Each cell refers to a chain. The song can contain up to 256 rows (`00-FF`).

Chain

A chain is an ordered list of phrases with optional transposition. Reusing a chain or phrase lets one edit repeated musical material once.

Phrase

A phrase row contains a note, instrument and FX columns. Notes use tracker notation such as `C-4`; an empty note does not retrigger the voice. Instrument changes are allowed on every row and a track may alternate freely between AY, Braids, Plaits and Sample instruments.

The FX selector only shows common FX plus the FX relevant to the instrument on that row. This prevents engine-specific commands from becoming one unmanageable global list.

7. Instruments

An instrument stores a name, type, optional transpose setting, table speed and engine-specific parameters. Use the Instrument Pool to browse all 128 slots, move instruments, copy/paste them and preview them.

Changing an instrument type resets the engine-specific data for that slot.

AY Classic, AY Plus and AY Sample

These are the original ChipNomad AY/YM engines. AY Classic exposes hardware-style tone, noise and envelope controls. AY Plus adds software oscillators and richer modulation. AY Sample reproduces a sample through AY-style volume levels and is distinct from the clean PCM Sample engine.

AY Quality in Settings affects only AY/YM rendering. It does not change Braids, Plaits or clean PCM sample quality. **Sample dithering** applies to AY Sample quantization.

Braids

Braids provides 47 synthesis models. Its main controls are:

- **Model**: oscillator/synthesis algorithm.
- **Timbre** and **Color**: model-dependent macro parameters.
- **Filter**: optional LP/BP/HP filter, 12 or 24 dB slope, cutoff and resonance.
- **ADSR**: attack, decay, sustain and release.

Braids terminology follows the original Mutable Instruments module: the audible result of Timbre and Color depends on the selected model.

Plaits

Plaits provides 24 engines in this build:

00–07	08–0F	10–17
VA VCF	ChooChooTracker User Manual Virtual Analog	Swarm
Phase Distortion	Waveshaping	Noise
6-op FM 1	2-op FM	Particle
6-op FM 2	Formant	String
6-op FM 3	Harmonic	Modal
Wave Terrain	Wavetable	Bass Drum
String Machine	Chord	Snare Drum
Chiptune	Speech	Hi-hat

The common Plaits macros follow Mutable's design:

- **Harmonic** controls harmonic relationships, balance or model choice inside an engine.
- **Timbre** generally moves from dark/sparse to bright/dense spectra.
- **Morph** explores a second timbral dimension.
- **Main/Aux** blends the main output with the engine's alternate output.

Their precise meaning is engine-dependent. For example, chord engines use them for chord type/inversion/waveform, physical models use them for material/excitation/decay, and drum engines use them for tone, character and decay.

The tracker adds an optional LP/BP/HP filter and ADSR after Plaits. One monophonic Plaits voice is allocated per tracker track.

PCM Sample

The clean Sample engine plays mono or stereo PCM independently of AY emulation.

- Tap **Sample** to load an uncompressed 16-bit PCM WAV.
- **Pitch** transposes by semitones (**-48** to **+48**).
- **Start** and **End** set normalized playback boundaries (**00–FF**).
- **Volume** is **00–FF** .
- The optional filter provides LP/BP/HP, 12/24 dB slope, cutoff and resonance.
- The ADSR shapes amplitude.

Unsupported WAV formats display a longer error message. Convert unusual files to PCM16 WAV before importing.

8. Modulation, Tables, Groove and Wavetables

Each instrument has four modulation slots. A slot can use ADSR, AHD or LFO behavior and targets a destination offered by that engine. Engine-specific destinations include Braids Timbre/Color, Plaits Harmonic/Timbre/Morph/Main-Aux and Sample playback/filter parameters.

Tables are small per-instrument sequences that can automate commands over time. Groove defines tick lengths used by phrase rows. Wavetables are used by compatible AY software oscillator modes.

9. Tracker FX

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FX have a three-letter command and a hexadecimal value. The in-app help panel gives a short description of the selected command.

Conditions and per-track speed

Conditions on the same row are combined with logical AND. A failed condition suppresses the row trigger.

FX	Value	Meaning
PRO	00-64	Absolute trigger probability from 0 to 100%
MOD	AB	Trigger on iteration A of B; for example 12, 22, 14, 34
SPD	00-10	Persistent playback speed for this track

MOD counters are local to the track and phrase. Invalid combinations, such as A greater than B, do not trigger.

SPD remains active until another SPD command changes it:

Value	Speed	Value	Speed
00	/7	09	Normal
01	/6	0A	×2
02	/5	0B	×4
03	/3	0C	×8
04	/32	0D	×3
05	/16	0E	×5
06	/8	0F	×6
07	/4	10	×7
08	/2		

Very high multipliers can exceed the resolution of the current tracker tick scheduler; validate ×4 and above with the intended groove on hardware.

Braids FX

FX	Meaning
BMD	Model
BTM	Timbre
BCL	Color
BCF	Filter cutoff
BRS	Filter resonance

Plaits FX

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FX	Meaning
PMD	Engine/model
PHA	Harmonic
PTM	Timbre
PMO	Morph
PAX	Main/Aux blend
PCF	Filter cutoff
PRS	Filter resonance

Sample FX

FX	Meaning
SPT	Signed semitone pitch
SST	Playback start
SEN	Playback end
SVL	Volume
SCF	Filter cutoff
SRS	Filter resonance

Engine parameter FX apply to their matching instrument type and are reset at the next note trigger unless stated otherwise. **SPD** is intentionally persistent.

Common and AY FX

The common group includes arpeggio, pitch, volume, retrigger, delay, note offset/kill, tick, table, groove, hop and song commands. AY-specific groups expose mixer, noise, hardware envelope, tone and software oscillator controls. Use the FX selection screen and its contextual help for the valid command set of the current instrument.

10. Project screen

Project provides Load, Save, New, Export and Manage commands, plus filename, title and author metadata.

- **Linear pitch** selects the pitch-table mode. For the current Braids integration, keep it **Off**: this is the validated setting and the default must produce correct octave tracking.
- **Tick rate** sets tracker timing and displays the corresponding BPM ($\text{tick rate} \times 60 / 24$).
- ChooChooTracker saves projects as **.cct**.

Save before changing instrument types or loading another project.

11. Settings

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- **Repeat delay / speed** tune held-button repeat.
- **Mix volume** controls final application output.
- **AY Quality** changes AY/YM emulation quality only.
- **Sample dithering** controls AY Sample dithering only.
- **Key mapping**, **Load font**, and **Edit color theme** customize the interface.
- **Quit ChooChooTracker** exits cleanly. No Menu+X shortcut is assumed.

12. Performance and troubleshooting

CPU

CPU cost depends on active engines and effects. Plaits physical models and Clouds Reverb are heavier than basic AY voices. Measure on the target console, especially with eight Plaits voices and both sends active. A CPU reading near 100% can cause crackles or missed audio deadlines.

No sound

Check the instrument number, track mute/solo state, track LVL, application Mix volume and the instrument envelope. For samples, verify that the original WAV is still at its saved path.

Wrong pitch

Use tonal material and verify several octaves. Keep Linear pitch Off for the currently validated Braids behavior. Percussive or noisy Plaits engines are not suitable tuner references.

Input feels wrong

Adjust Repeat delay and Repeat speed in Settings. If a single press moves twice, verify that only one physical control is mapped to that direction and report the exact screen and shortcut.

13. Credits and licensing

ChooChooTracker is a fork of ChipNomad and retains its MIT licensing approach. Braids, Plaits, Clouds DSP and stmllib code are derived from Mutable Instruments' open-source releases under their applicable MIT notices. See the packaged license files for exact attribution.

The descriptions of Mutable synthesis behavior in this manual are adapted to ChooChooTracker's controls from the original Mutable Instruments Braids, Plaits and Clouds documentation included in the repository's

inspirations directory.